



Waterboard Management System

System Administrator Guide

Version: 1.0

Prepared for:

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Revision History

Version	Date	Description of Change	Author
1.0	July 11, 2026	Initial release of the Administrator Guide.	Admin Team

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1 1. Introduction

Purpose

This document provides system administrators with the necessary instructions to manage, monitor, and configure the Waterboard Management System.

Audience

This guide is intended for IT administrators, network operators, and billing managers responsible for the daily operation of the smart water meter network.

Scope

This manual covers software platform navigation, user management, device monitoring, billing configuration, and basic system troubleshooting. Detailed electronic schematics and physical maintenance procedures are covered in the separate *Hardware Maintenance Manual*.

Prerequisites

- A modern web browser (Google Chrome, Mozilla Firefox, or Microsoft Edge).
- Authorized administrative credentials and the current system authorization code.
- A stable internet connection.

2 2. System Overview

The Waterboard Management System is a centralized platform that aggregates telemetry data from field-deployed smart meters.

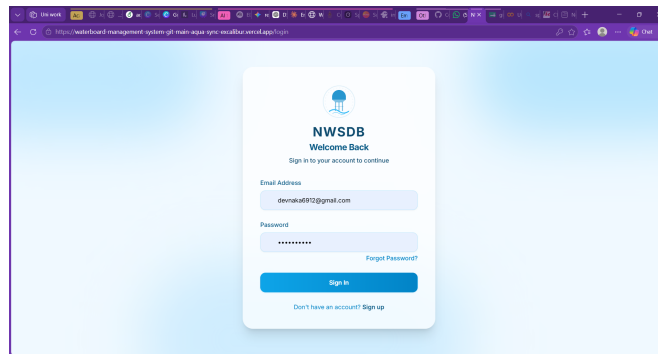
Each deployed water meter contains:

- A core processing microcontroller.
- A flow sensing unit (magnetic rotation detection).
- A GSM communication module for remote data transmission.
- A long-life, industrial-grade battery.

The admin portal continuously processes this hardware data into actionable billing and operational metrics.

3 3. Login

Purpose: Authenticate and securely access the administrator portal.



Admin Login Interface

Procedure:

1. Navigate to the admin portal URL in your web browser.
2. Enter your authorized administrator **Email**.
3. Enter your **Password**.
4. Enter the mandatory authorization code: **WATER2026**.
5. Click **Login**.

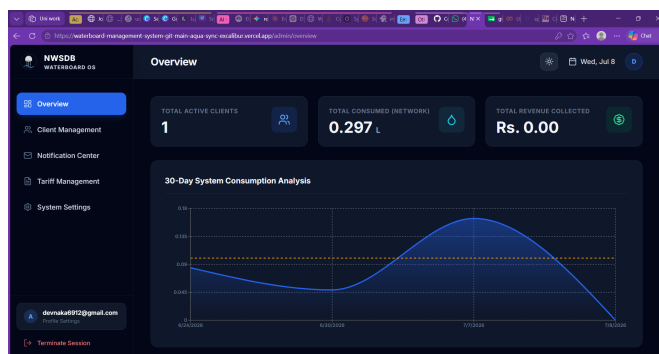
Expected Result: The system authenticates the credentials and redirects to the Admin Dashboard.

Notes:

- Failed attempts exceeding 5 tries will temporarily lock the account for 15 minutes.

4 4. Dashboard

Purpose: Provide a high-level overview of system health and key metrics.

*Dashboard Overview***Procedure:**

1. Log in to the system. The Dashboard is the default landing page.
2. Review the **Global Metrics** panel for total active clients and revenue.
3. Review the **System Status** panel for offline hardware warnings.

5 5. Client Management

Purpose: Manage customer accounts, update profiles, and review individual consumption.

Procedure:

1. Navigate to **Client Management** using the left sidebar.
2. Search for the client using the top search bar (by name or meter ID).
3. Click **View Details** next to the desired client.
4. Edit the required fields (e.g., contact information, account status).
5. Click **Save**.

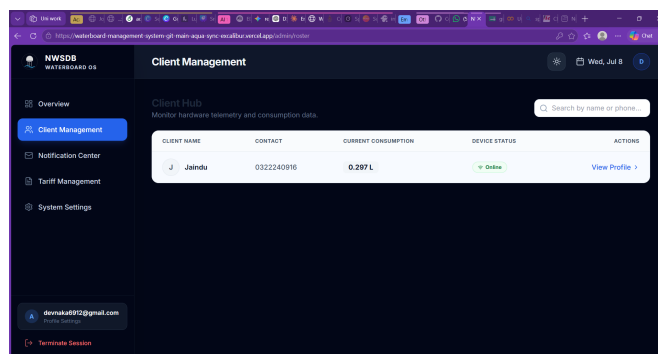
Expected Result: The client profile is updated successfully and reflects immediately in the database.

Notes:

- Email addresses must be unique across the system.
- Deleted accounts cannot be recovered. Use "Suspend" instead of deleting where possible.

6 6. Device Management

Purpose: Monitor the operational status and data synchronization of field-deployed water meter assets.

*Device Management Interface***Procedure:**

1. Navigate to **Devices** in the primary dashboard sidebar.
2. Select the desired meter ID from the fleet register table.
3. Validate the following interface metrics:
 - **Total Consumption:** The cumulative water usage recorded by the hardware.
 - **Last Handshake Timestamp:** Indicates the exact date and time the system last received a telemetry payload from the GSM module.

Notes:

- Telemetry data updates on an hourly cycle. If the Last Handshake Timestamp exceeds 24 hours, the system marks the node state as "Offline," indicating a transmission failure.

7 7. Tariff Management & Billing

Purpose: Configure pricing structures and unit rates for water consumption.

Procedure:

1. Navigate to **Tariffs & Billing**.
2. Select the billing region or tier you wish to modify.
3. Update the **Unit Rate** (per Liter) or **Fixed Charge** fields.
4. Click **Apply to Network**.

Expected Result: The new pricing structure will be applied to the next billing cycle. Current open invoices will not be affected.

8 8. Bulk Notifications

Purpose: Broadcast alerts, maintenance updates, or policy changes to clients.

Procedure:

1. Navigate to **Notifications**.
2. Click **Create New Message**.
3. Select the target audience (All Clients or Specific Region).
4. Type the message subject and body.
5. Click **Send Notification**.

9 9. System Settings

Purpose: Manage global application parameters and administrator profiles.

Procedure:

1. Navigate to **Settings**.
2. Update application preferences or click **Admin Profile** to change your password.
3. Click **Save Changes**.

10 10. Troubleshooting

Diagnostic Procedures & Core Issues

Procedure: Offline Device Diagnostics

Because the smart meters transmit minimal payload data to conserve power, remote telemetry for battery and signal strength is not available on the dashboard. Diagnostics require isolating the fault using timestamp data followed by a localized field inspection.

Core Issue: Device Status shows as 'Offline' (No data received for ≥ 24 hours).

Step 1: Verify the Data Interruption Time

1. Navigate to **Device Management** and check the **Last Handshake Timestamp**.
2. If multiple devices in the same geographic region went offline at the exact same timestamp, the issue is likely a regional cellular network outage rather than a hardware failure.
3. If only a single device is offline, flag the unit for a physical field inspection.

Step 2: Field Inspection - Power Status

1. Dispatch a technician to the physical meter location.
2. Press the local **wake button** on the front face of the meter.
3. **Result A:** If the digital display remains blank, the internal power cell has been exhausted. The unit requires a battery replacement.
4. **Result B:** If the display wakes up and shows the Liters consumed, the hardware is powered and the issue is related to the GSM transmission.

Step 3: Field Inspection - GSM Signal Shielding

1. If the meter has power but is not transmitting, inspect the installation environment.
2. Ensure the client has not placed heavy metal enclosures, dense objects, or RF-shielding materials around the meter that would block the GSM module from reaching the cellular tower.

Expected Result: The administrator successfully categorizes the failure as a network outage, a dead battery, or RF shielding, allowing for accurate maintenance dispatch.

11 11. Frequently Asked Questions

Q: Can I recover a deleted client profile?

A: No, deleted profiles are permanently removed from the active database for privacy compliance. Always use the 'Suspend' feature unless a permanent deletion is explicitly required.

Q: How often do the meters transmit data?

A: Under standard configuration, meters transmit telemetry data hourly to optimize battery life while providing near real-time billing data.

12 12. Contact & Support

For advanced technical assistance, database architecture inquiries, or hardware replacements, please contact the development team:

- **Department:** Electronic and Telecommunication Engineering (ENTC)
- **Institution:** University of Moratuwa
- **System Documentation:** Refer to the Internal Backbone Network Guide for routing and server configurations.